

90. What is EDM used for on Calibration blocks?

- a. Creating notches
- b. Drilling side drilled holes
- c. Shaping the block
- d. Creating flat bottom holes

N261 top (standards)

91. When soldering what is the dimension a pipe has to be moved out of the socket during fit up?

- a. 1/16 inch
- b. 1/8 inch
- c. 1/2 inch
- d. 1/4 inch

B31.1, para 127.3 E; VT Hdbk 2nd ed page 171 and 174 (Figures); VT 3rd ed, 217, left, 1st

92. Two discontinuities commonly found in forgings and castings?

- a. Bursts and blowholes
- b. Laps and seams
- c. Laps and stringers
- d. Tungsten inclusions and blowholes

Old 85, left, 3rd

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PART C – NDT METHODS

PENETRANT

1. The natural forces relied upon to the greatest extent in liquid penetrant testing is called:
 - a. Gravity
 - b. Boyle's Law
 - c. Capillary action
 - d. Blotting quotient
2. Interpretation of penetrant indications as required by most specifications is based on:
 - a. Length and width of the discontinuity opening
 - b. Area the discontinuity opening
 - c. Length and width of the bleed out
 - d. Depth of discontinuity opening
 - e. Contrast to the penetrant with developer

ASNT Hdbk 3rd ed, page 349, left col, 4th para and ASME code contradict

3. Liquid penetrant tests can be used to detect:
 - a. Internal porosity in castings
 - b. Corrosion wall thinning in pipes and tubes
 - c. Fatigue cracks
 - d. Carbon content of steels
 - e. Hardness variation in stainless steel forgings

4. When using fluorescent penetrants, a black light source is desirable at the wash station:
 - a. So the inspection can be made before the part dries
 - b. To observe the penetrant bleeding out of the discontinuities
 - c. To check the effectiveness of the wash cycle
 - d. To observe the intensity of the developer
 - e. To assure removal of all penetrant
5. Water washable liquid penetrants differ from post-emulsification in that water washable penetrants:
 - a. Can only be used on aluminum test specimens
 - b. Need not be removed from services prior to development
 - c. Have a soapy base
 - d. Do not need the application of emulsifier before rinsing
6. Which of the following is not a good practice when performing a penetrant test?
 - a. Applying emulsifier by dipping the part in emulsifier
 - b. Applying developer by spraying the part with developer
 - c. Removal of water washable penetrant with the water spray
 - d. The applying emulsifier with a brush
7. Which of the following is an advantage of fluorescent penetrants over visible dye penetrants?:
 - a. The inspection can be carried out in a well lighted area
 - b. Small indications are more easily seen
 - c. They can be used where contact with water is objectionable
 - d. No special lighting is required

8. Which of the following is an advantage of visible dye over fluorescent penetrants?:
- a. Small indications are more easily seen
 - b. They can be used in anodized and chromate surfaces
 - c. They make less background on rough surfaces
 - d. No special lighting is required
9. Which of the following is an advantage of solvent removable methods over water washable?:
- a. No special lighting is necessary during inspection
 - b. They provide a quicker penetration of small openings
 - c. Small indications are more easily seen
 - d. They can easily be performed in the field and remote areas
10. Developer should always be:
- a. Highly fluorescent
 - b. Applied wet
 - c. Colorless
 - d. Evenly applied

11. When performing liquid penetrant examination at temperatures below 40°F, which of the following apply?
- a. Extend the dwell time
 - b. Compare the materials to be used under the anticipated low temperatures conditions
 - c. Check for penetrant stratification
 - d. Avoid use of water processing

12. Penetrants may be used for detecting leaks:

- a. If surfaces are clean
- b. Both sides are accessible
- c. In dilute form during hydro test
- d. Effectively by using fluorescent tracers
- e. In all the above

13. Of the following, which would be the principal factor in selecting a fluorescent penetrant over a visible penetrant for a particular testing situation?

- a. The part had anodic or chromate treatment
- b. Resolution of fine indications below the visual threshold is required
- c. The part had been cleaned in either an alkaline or acid bath
- d. Resolution of fine indications that are above the visual threshold is required
- e. All of the above may influence the choice

14. What does it mean when they say that PT is a homogenous method?

- a. Agitation is required to keep the penetrant constituents mixed
- b. No agitation is required to maintain penetrant constituents in solution – penetrant should be uniform
- c. Penetrant constituents separate when left standing
- d. Penetrant materials are inconsistent

MAGNETIC PARTICLE

15. Magnetic particles are attracted to discontinuities in the part being magnetized by:

- a. The flow of voltage across the discontinuity
- b. Electrostatic attraction
- c. Gravitational attraction
- d. Magnetic flux leakage
- e. Magnetism of like poles

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16. In the continuous method, excess dry particles should be removed from the surface of the part being examined:

- a. When the magnetizing current is off
- b. When the magnetizing current is on
- c. Makes little or no difference
- d. After a short holding time
- e. Excess dry particle should be left on the surface

17. The magnetic lines of forces are in what direction in relation to the direction of the magnetizing current?:

- a. Parallel
- b. At right angles
- c. A 45° angles
- d. Many directions
- e. Reversing 180° sixty times every second

18. The closer a discontinuity is to the surface:

- a. The more diffuse becomes the particle pattern
- b. The more distinct becomes the particle pattern
- c. The less serious it is
- d. The less intense become the flux leakage field
- e. The more difficult it becomes to detect

19. In magnetic particle inspection the accumulation of particles on the surface of the part is called:

- a. Discontinuity
- b. Indication
- c. Defect
- d. Flaw
- e. Imperfection

20. Wet fluorescent magnetic particle inspection is sometimes used in preference to colored magnetic particles:

- a. To increase speed of inspection
- b. To inspect parts with rough surfaces
- c. To obtain maximum sensitivity
- d. To take advantage of the effect of fluorescent lights
- e. To decrease to sensitivity

21. Why are large soft contact areas such as lead or copper braid used for surfaces or headstocks? :

- a. To increase the contact area and reduce the possibility of burning the part
- b. Because of their low melting points
- c. Because they help heat metal, thus facilitating magnetic induction
- d. To increase the contact area and flux density

22. If one inch and two inch diameter bars were magnetized by passing the same current through them, the magnetic fields would be:

- a. The same for both
- b. Stronger in the two inch diameter bar
- c. Weaker in the 1 inch diameter bar
- d. Stronger in the 1 inch diameter bar

23. The best method for inspection of a large casting is:

- a. Multidirectional magnetization
- b. A central conductor
- c. Prod inspection in two 90° opposing directions
- d. Direct magnetization using an AC yoke

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24. Materials which are strongly attracted to a magnet are called:

- a. Magnetized
- b. Nonmagnetic
- c. Ferromagnetic
- d. Polarized

25. Which of the following is the most effective method of demagnetization?:

- a. Alternating current
- b. Direct current
- c. Half-wave rectified alternating current
- d. Direct current surge

26. Which field is more difficult to demagnetize?

- a. Circular
- b. Longitudinal
- c. Multidirectional
- d. Vectored

27. Which method would employ used of a Hall meter?

- a. Liquid penetrant
- b. Magnetic Particle
- c. Eddy Current
- d. Leak Testing

28. How is HWDC is different than FWDC?

- a. More surge from HWDC
- b. Particles have more mobility with HWDC
- c. The current reverses direction every half cycle

29. X-rays travel at a velocity:

- a. Dependent upon the KV used
- b. Equal to the speed of light
- c. Proportional to the square of the distance
- d. Determined by the distance traveled

30. To what is the primary difference between x-rays and gamma rays of the same energy?:

- a. Wavelength
- b. Frequency
- c. Velocity
- d. Origin
- e. Energy level

31. The inverse square law as applied to radiation states that:

- a. Radiation intensity varies inversely as the square of the time spent near the source
- b. Radiation intensity varies proportionally with the distance from the source
- c. Radiation intensity varies inversely to the square of the distance from the source
- d. Radiation intensity has inversely proportional to the exposure time

32. An x-ray tube with a small focus spot is considered better than one with a large focal spot when it is desired to obtain:

- a. Greater penetrating power
- b. Less contrast
- c. Better definition
- d. Greater film density
- e. A shorter exposure time

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33. A densitometer is:

- a. A meter used to measure x-ray and intensity
- b. An instrument for measuring film density
- c. A meter used to measure tube current
- d. A survey device to detect radiation

34. A penetrameter is used to measure:

- a. The size of discontinuities that can be detected in the part
- b. The density of the film
- c. The amount of film contrast
- d. The quality of the radiographic technique
- e. The graininess of the film

35. The accidental movement of the specimen or film during exposure will:

- a. Produce the radiograph with poor contrast
- b. Make it impossible to detect large discontinuities
- c. Result in unsharpness of the radiograph
- d. Result in a fogged radiograph
- e. Cause the radiograph to be overexposed

36. The difference between the densities of two areas of a radiograph is called:

- a. Radiographic contrast
- b. Subject definition
- c. Film characteristic curves
- d. Detail
- e. Density gradient

37. Radiographic films with large grain size:

- a. will produce radiographs with better definition than film with small grain size
- b. have slower speeds than those with a relatively small grain size
- c. have higher speeds than those with a relatively small grain size
- d. will take longer to expose properly than film with relatively small grain size

38. In order for a radiograph to have a penetrameter sensitivity of 2 - 2T or better:

- a. The radiograph has to differentiate a 2% difference in specimen thickness
- b. When the radiograph shows the 2T hole in a penetrameter, where the penetrameter is 2% of the thickness of the specimen
- c. The radiograph must be able to distinguish a discontinuity with a length equivalent of 2% of the specimen thickness
- d. The 2T hole must be observed in a penetrameter which is 20% of the specimen thickness

39. Which of the following factors will reduce radiographic geometric unsharpness?:

- a. The focal spot should be as small as possible
- b. The source to object distance should be as great as possible
- c. The beam of the radiation should be as nearly perpendicular as possible
- d. All of the above
- e. A. and B. above

40. A qualitative term used to indicate the size of the smallest detail which can be seen on a radiograph is:

- a. Radiographic sensitivity
- b. Radiographic definition
- c. Radiographic contrast
- d. Subject contrast

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41. Film contrast is dependent on which of the following factors?:

- a. The type of screens used
- b. The density of the radiograph
- c. The type of film used
- d. All of the above

42. A satisfactory radiographic exposure is obtained using a 5 Ci source and 10 minutes. If a 20 Ci source was used, how long would film need to be exposed to get the same degree of film darkening?

- a. 40 minutes
- b. 10 minutes
- c. 5 minutes
- d. 2.5 minutes
- e. 2 minutes

43. After a radiographic source has aged 3 half lives, by what factor will an exposure need to be changed in order to get the same degree of film darkening?

- a. 3 times greater
- b. 6 times greater
- c. 8 times greater
- d. 1/8 as long
- e. Half as long

44. A radiation level of 100 mR/hr is detected. The half value layer thickness for lead when Cobalt 60 is used as a source is 0.5 inch. If 2 inches of lead is used for shielding, what will the radiation exposure rate be on the other side of the shielding?

- a. 25 mR/hr
- b. 6.25 mR/hr
- c. 400 mR/hr
- d. 50 mR/hr

45. How long should an inspector wait to handle a part after exposure to Co 60?

- a. The part may be picked up immediately after exposure
- b. 1 minute
- c. 5 minutes
- d. 10 minutes

46. How long should an inspector wait to handle a part after exposure to X-rays?

- a. The part may be picked up immediately after exposure
- b. 1 minute
- c. 5 minutes
- d. 10 minutes

47. If 200 mR/hr is detected at 25 ft during an X-ray exposure, what would the exposure rate be at 125 ft?

- a. 8 mR/hr
- b. 40 mR/hr
- c. 5000 mR/hr
- d. 1125 mR/hr

Inverse square law

48. What is the normal configuration for double-loading a cassette when shooting with IR-192?

- a. Film, screen, film, screen
- b. Screen, film, film, screen
- c. Film, screen, screen, film
- d. Screen, film, screen, film

49. If a film is developed for 5 minutes in 68 F developer, how long is it to be in the fixer at 68 F?

- a. Double clearing time
- b. 5 minutes plus double the clearing time
- c. 5 minutes + Clear time
- d. 10 minutes + Clear time

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ULTRASONICS

50. Which of the following materials is most likely to cause the greatest amount of attenuation over a given distance?:

- a. Forging
- b. Coarse-grained casting
- c. Extrusion
- d. Fine-grained material
- e. Rolled steel plate

51. In the a-scan presentation, the signal position along the sweep is an indication of:

- a. The distance between the discontinuity and the surface of the material being examined
- b. The amount of interference offered by the area of the discontinuity
- c. The type of wave used
- d. The wavelength of the ultrasonic beam
- e. The length of the reflector

52. Which of the following is an advantage of contact testing over immersion?:

- a. Greater sound transmission into the test part
- b. Higher frequencies may be used
- c. Better resolution
- d. Improved sensitivity
- e. Faster scanning is possible

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53. Which of the following is a disadvantage of contact testing over immersion?

- a. Decreased attenuation
- b. Portability
- c. Difficulty in maintaining uniform acoustical coupling
- d. Confined to high frequency testing only
- e. Improved resolution

54. If the major dimensions of a discontinuity in a 6" thick aluminum plate lie parallel to the sound entry surface at a depth of 3", it will be best detected by:

- a. A compressional beam test
- b. An angle beam test
- c. A surface wave test
- d. A Lamb wave test
- e. Through transmission

55. Increasing frequency in ultrasonic testing:

- a. Increases depth of penetration
- b. Produces skin effect
- c. Increased sensitivity
- d. Necessitates use of special transducers
- e. Decreases sensitivity

56. In ultrasonic testing, a couplant must be used between the test object and the transducer because:

- a. The transducer must be lubricated in order to function
- b. The lubrication of the test object surface facilitates movement of the transducer over the surface
- c. Ultrasonic sound waves are rapidly attenuated in air
- d. Lubrication accelerates the passage of sound waves through objects
- e. Mode conversion is not possible in a liquid

57. The velocity of sound is highest among the materials listed below in:

- a. Air
- b. Water
- c. Aluminum
- d. Stainless steel
- e. Lead

58. Which ultrasonic test frequency would provide the best penetration in a 12-inch thick specimen of coarse-grained steel?:

- a. 2.25 megahertz
- b. 1 megahertz
- c. 5 megahertz
- d. 10 megahertz

59. In contact testing, discontinuities near the sound entry surface cannot always be detected because of:

- a. The far-field effect
- b. Attenuation
- c. The dead zone
- d. Refraction
- e. The near zone

60. In ultrasonic examination, the ability to resolve two discontinuities at different depths would be improved by:

- a. Decreasing frequency
- b. Increasing pulse length
- c. Increasing the amplitude of the initial pulse
- d. Increasing frequency

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61. Which of the following material characteristics will affect the acoustical attenuation in the material?

- a. Grain size
- b. Material elasticity
- c. Crystalline structure
- d. All of the above

62. The longitudinal velocity of sound in a material is:

- a. A function of density only
- b. A function of density, Young's Modulus and Poission's Ratio
- c. Directly proportional to the incident beam angle
- d. Determined by test frequency

63. Ultrasonic waves in test materials are:

- a. Electromagnetic waves
- b. Low voltage electrical fields
- c. Discontinuous waves
- d. Mechanical vibrations

64. In UT two transducers used from each side of the plate

- a. contact method
- b. shearwave
- c. pitch catch
- d. through transmission

65. Eddy current testing relies on the principle of:

- a. Magnetostriction
- b. Electromagnetic induction
- c. Piezoelectric energy conversion
- d. Magnetomotive Force

66. Eddy Current testing is the best NDT method for detecting:

- a. Laminations in plates
- b. Lack of weld penetration in thick stainless steel pipe
- c. Internal defects in small diameter tubing
- d. Lack of bond in plating
- e. All of the above

67. The three major specimen parameters which influence eddy current testing are:

- a. Electrical conductivity, frequency, and material geometry
- b. Density, permeability and frequency
- c. Electrical conductivity, permeability and material geometry
- d. Thermal conductivity, electrical conductivity and permeability

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68. A term used to describe the effect observed due to a change in the electromagnetic coupling between a test specimen and a probe coil when the distance of separation between them is varied is:

- a. Fill-factor
- b. Edge Effect
- c. End Effect
- d. Lift Off

69. What is the meaning of magnetic permeability?:

- a. a measure of ability of material to conduct a magnetic field
- b. a measure of gap between the probe and surface being tested
- c. a measure of ability of material to conduct eddy currents
- d. a measure of ability of a stationary permanent magnetic to induce currents in a conductor

70. Eddy current NDT method can be used to detect:

- a. Discontinuities in thin material
- b. Surface discontinuities in thick material
- c. Variations in electrical resistivity
- d. All of the above

71. Tubing is generally inspected using:

- a. U-shaped coils
- b. Gap coils
- c. Internal coils
- d. None of the above

72. When a rod is placed in an encircling type of coil, the density of eddy currents will be the greatest:

- a. At the surface
- b. At the center
- c. Midway between the surface and the center
- d. None of the above

73. In eddy current testing, current is induced in a conductor from a:

- a. Stationary magnetic field from a strong magnet
- b. Coil carrying direct current (DC)
- c. Coil carrying alternating current (AC)
- d. Current flowing directly through the sample by electrical contact

74. The thickness of nonconductive coatings on a conductive base can be most simply measured by:

- a. Observing the lift-off effect caused by the coating
- b. Testing both sides of the specimen
- c. Varying the current in the test coil
- d. Varying the test frequency over a given range during the test

75. A decrease in conductivity is equivalent to:

- a. an increase in permeability
- b. an increase in resistivity
- c. a decrease in permeability
- d. a decrease in resistivity

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76. Flux Leakage inspection can be applied to:

- a. Ferromagnetic and nonmagnetic materials
- b. Nonmagnetic materials only
- c. Ferromagnetic materials only
- d. Nonconductors only

77. The magnetic field generated by eddy currents induced in a test specimen

- a. Opposes the magnetic field that induced the eddy currents
- b. Reinforces the magnetic field that induced the eddy currents
- c. Cancels the magnetic field that induced the eddy currents
- d. Has no effect on the magnetic field that induced the eddy current

78. Increasing test coil current will:

- a. Increase depth of penetration
- b. Decrease depth of penetration
- c. Not change depth of penetration
- d. Increase lift-off

79. A term used to define the phenomenon where at high frequencies, the eddy current flow is restricted to an extremely thin outer layer of the conductor is:

- a. Skin effect
- b. High frequency filtration
- c. Low frequency
- d. Any one of the above

80. Eddy current would be best suited for _____.

- a. Measuring debond of a plating
- b. Determining heat treat condition of 17-50 aluminum alloy
- c. Finding circumferential flaws using a bobbin coil
- d. Cracks in plastic

81. Increasing eddy current frequency has what effect on depth of penetration?

- a. circular
- b. increases depth of penetration
- c. decreases depth of penetration
- d. no change in depth of penetration

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PART D – MISCELLANEOUS NDT METHODS

1. A helium mass spectrometer is used in which of the following NDT method(s):

- a. Penetrant
- b. Magnetic Particle
- c. Acoustic emission
- d. Leak testing

2. Best NDE method to check a Boom under LOAD?

- a. Vibrations
- b. Acoustic emission
- c. Radiography
- d. Ultrasonic

3. Which would most likely be found by visual inspection?

- a. Lack of fusion
- b. Lack of penetration
- c. Slag inclusion
- d. Undercut

4. Condensation is on the inside wall of a building. If this evaporates this would have what effect on the temperature of the wall?

- a. Cooling
- b. Warming
- c. No effect

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5. What method would be used to show whether or not a plastic gasket was properly placed or present inside a steel case?

- a. Ultrasound
- b. Radiography
- c. Eddy current
- d. Neutron radiography

6. What type of film is used for neutron radiography?
- Specially formulated NR film
 - X-ray film
 - Polaroid
 - Film is not used
7. What method would be used to identify missing solid fuel in a rocket or explosive?
- Radiography
 - Penetrant
 - Eddy current
 - Neutron radiography
8. What would cause a change in material velocity within a material?
- Mode conversion
 - Vibration
 - Higher frequency
 - Lower frequency
9. What method would be used to leak test a 1200 psi pressure vessel having access to both sides?
- Penetrant
 - Halogen Diode
 - Light Static Pressure
 - Mass Spectrometer.

ASNT NDT Hdbk Vol 10, 3rd ed, page 146-147 and Fig 5

10. DELETED

11. Where does the energy come from during acoustic emission?

- a. Transducer
- b. Beam
- c. Current
- d. Part

12. During AE, the Kaiser effect is useful in what regard?

- a. No emission should occur until the previous loads have been exceeded, which would separate mechanical noise from growing discontinuities
- b. Electrical noise from mechanical noise
- c. Electrical noise from continuous emission
- d. Electrical noise from growing discontinuities

13. The most common range for acoustic emission is :

- a. 10-15 KHz
- b. 100-300 KHz
- c. 500-800 KHz
- d. 1-5 MHz

14. Acoustic emission readily detects:

- a. Leaks
- b. Plastic deformation
- c. Growing cracks
- d. All of the above

15. The main reason for using neutron radiography in place of X-ray is:

- a. Lower cost
- b. Higher resolution
- c. The ability to image objects and materials not possible with X-ray
- d. It is simpler than X-ray

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16. How is the film exposed during neutron radiography?

- a. Neutrons ionize the film
- b. Neutrons render the film more radioactive where the material is thin providing contrast between thick and thin areas
- c. Neutrons bombard a screen causing emission of electrons, gamma rays or light which subsequently ionizes radiographic film
- d. Neutrons cause the screen material to become radioactive and film is subsequently placed in a cassette with the screen where it is exposed to gamma rays through radioactive decay of the screen material
- e. c and d

17. The most sensitive leak test method is:

- a. Bubble
- b. Pressure change
- c. Halogen diode
- d. Helium Mass Spectrometer

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18. A vacuum box is used for which LT technique.

- a. Bubble
- b. Pressure change
- c. Halogen diode
- d. Mass spectrometer

ASNT NDT Hdbk Vol 10, 3rd ed, page 154-155

19. In a fiber optic borescope how is the image transmitted _____.

- a. From lens to lens
- b. Fiber to fiber
- c. Lens to fiber to eye piece
- d. Lens to fiber to lens to fiber

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20. How is depth of field affected by magnification?

- a. Depth of field increases with increased magnification
- b. Depth of field decreases with increased magnification
- c. Depth of field is not affected by magnification
- d. Depth of field increases and then decreases as magnification increases

21. What method would be best suited to detect the presence of solid rocket fuel inside a 2" thick steel casing?

- a. MT
- b. UT
- c. ET
- d. NRT

22. During infrared inspection of a building from the outside, condensation is developing on the inside walls. What effect will this have on the temperature of the wall?

- a. Decreases temperature upon evaporation
- b. Increases temperature upon condensation
- c. Have no effect on temperature

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23. What effect does increasing magnification have on depth of field (DOF) and field of view (FOV)?

- a. Decreases DOF and FOV
- b. Increases DOF and FOV
- c. Decreases DOF and increases FOV
- d. Increases DOF and decreases FOV
- e. Has no effect on either

24. A dense fiberglass shaft is under a heavy stress load. What form of NDT would be best to determine if any cracks are forming?

- a. RT
- b. AE
- c. MT
- d. UT

25. There is an AE signal at 150 KHz with an acoustic velocity in the material of 3000 m/s. What is the wavelength?

- a. 0.05 mm
- b. 20 mm
- c. 20000 m
- d. 0.05 m

$V=f \times \lambda$, where λ is the wavelength. Convert KHz to Hz, and m/s to mm/s.

26. A helium mass spectrometer is used for _____?

- a. Leak Location
- b. Detecting Leaks
- c. A & B
- d. Vacuum box

LT Hnbk, 3rd ed, page 320; 283 & 613 def of vac box; ASNT NDT Hdbk Vol 10, 3rd ed, page 146

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27. A fiber-optic Borescope contains:

- a. Steel rods holding the lenses together
- b. Polished glass rods
- c. Clad glass fibers
- d. Copper rods

28. What is the force that drives thermal conduction?

- a. Gas or liquids transferring heat between objects
- b. Pressure difference
- c. Emission of electromagnetic radiation as temperature increases
- d. Temperature difference

ASNT NDT Hdbk Vol 10, 3rd ed, page 179

29. Weld with access to both sides with 15 psi pressure

- a. current flow
- b. pressure difference

LT Hdbk 306; 295, LC, 4th para Bubble testing = unless structurally limited or otherwise specified, the minimum pressure differential between the pressure of the test object and the external pressure at the greatest depth in the immersion liquid should be 15 psi (100 kPa). Repeated in 298, LC, 2nd para

30. Which leak test technique would be chosen for a 15 psi pressure vessel where no leaks are allowed?

- a. Fluorescent penetrant
- b. Bubble test
- c. Pressure change
- d. Mass spectrometer

Psig is pressure above atmospheric (14.7 psi)

31. What would Acoustic emission be used to evaluate?

- a. Size flaws in a weld
- b. Porosity in a casting
- c. Boom lift
- d. Identify bursts in forgings

32. Which of the following cannot be determined using a helium mass spectrometer?

- a. Presence of a leak
- b. Location of a leak
- c. Size of a leak
- d. None of the above

33. A 150 KHz transducer is used for acoustic emission. This frequency is _____.

- a. Below the audible hearing range
- b. In the ultrasonic range
- c. Above the ultrasonic range
- d. In the middle of the audible hearing range

34. Yaw, pitch and roll are important for _____ ?

- a. Digital radiographic interpretation and evaluation
- b. Mechanically controlling position of a probe
- c. Acoustic Emission
- d. Leak Testing utilizing a helium mass spectrometer

35. Increasing frequency during an eddy current test will _____ depth of penetration.

- a. Decrease
- b. Increase
- c. Have no effect on
- d. Increase and then decrease

36. Energy measured during acoustic emission originates from _____.

- a. The transducer
- b. The part
- c. The part and transducer
- d. Seismic activity

37. During immersion ultrasonic inspection, the amount of sound transmitted into the material is _____.

- a. Less than the amount of sound reflected
- b. Greater than the amount of sound reflected
- c. Equal to the amount of sound reflected
- d. Transmitted sound is refracted at 120 degree angle relative to the angle of incidence

38. Of the following which is most often used, but is not a recognized NDT method?

- a. Shearography
- b. Concrete inspection
- c. Metrology
- d. Holography

Reported that coating was not available - shearography

39. A Hall Effect probe would be used for which of the following methods:

- a. Eddy current and magnetic particle
- b. Radiography and eddy current
- c. Radiography and ultrasonic
- d. Magnetic Particle and Magnetic Flux Leakage

40. A measurement tool to identify contour and pitch of threads is called a _____.

- a. Ultrasonic thread probe
- b. Thread profile gage
- c. Conductivity profile gage
- d. Ruler

41. The SI unit for the range of wavelengths in the electromagnetic spectrum used for visual inspection is _____.

- a. Lux
- b. Lumen
- c. nm
- d. Angstrom

ASNT VT Hdbk, 3rd ed, page 35, Table 9; ASNT NDT Hdbk Vol 10, 3rd ed, page 20-21, Table 6

42. What characteristic do AE and LT have in common that distinguishes them from other NDT methods?

- a. Transducer
- b. Electronic probe
- c. Energy originates from part
- d. Energy is transmitted through the part

ASNT LT Hdbk 3rd ed, page 8, left col (direct test); ASNT NDT Hdbk Vol 10, 3rd ed 416 (AE) and 432 (AE for LT)

43. Which of the following materials has the greatest electrical conductivity in %IACS?

- a. Silver
- b. Copper
- c. Aluminum
- d. Stainless Steel.

Page 66, Table 6-5; ASNT ET Hdbk, 3rd ed, page 331, Fig 11; ASNT NDT Hdbk Vol 10, 3rd ed, page 497 Figure 3.

44. What are the safety precautions while using a tracer gas with LT?

- a. Flammability
- b. Asphyxiation
- c. Physiological effects
- d. Pressure vessel explosion
- e. All of the above

ASNT LT Hdbk 3rd ed, page 123, left col

45. A source produces 35 mR/hr at 35 feet, how much energy will it produce at 180 feet?

- a. 6.8 mR/hr
- b. 180 mR/hr
- c. 238 mR/hr
- d. 1.3 mR/hr

Inverse Square Law; ASNT NDT Hdbk Vol 10, page 218

46. Magnetic lines of force are strongest _____?

- a. After applying the field
- b. Before applying the field
- c. At the point the field reverses direction
- d. While the field is being applied

ASNT MT Hdbk, 3rd ed, page 64, left col, 3rd para; ASNT NDT Hdbk Vol 10, 3rd ed, page 300

47. You have 1 inch thick material and you are using 5 inch spacing between prods. What would be the proper current in amperes?

- a. 50 amps
- b. 150 amps
- c. 300 amps
- d. 625 amps

ASNT MT Hdbk, 3rd ed, page 113, left col, 4th para

48. Lower frequency during eddy current will:

- a. reduce penetration
- b. increase penetration
- c. focus eddy current
- d. not affect penetration

ASNT ET Hdbk, 3rd ed, page 130, rt col

49. A change in acoustic velocity will occur with ____.

- a. An increase in frequency
- b. A decrease in frequency
- c. Mode conversion
- d. Reflection

CT-6-4, page 2-18

50. Ultrasonic sensitivity increases when transducer element ____.

- a. Thickness increases
- b. Thickness decreases
- c. Diameter increases
- d. Turns decreases

CT-6-4, page 3-20; ASNT NDT Hdbk Vol 10, 3rd ed, page 339, left col

51. What is the level of sensitivity for penetrant testing for extremely tight cracks?

- a. Low
- b. Medium
- c. High
- d. Ultra high

ASNT PT Hdbk, 3rd ed, page 39, left col, last and

52. What would you do if you notice that during rinsing of a post-emulsifiable penetrant it is removed VERY FAST?

- a. Increase penetrant
- b. Increase hydrophilic
- c. Increase dwell time
- d. Reduce emulsifier concentration or dwell time

ASNT PT Hdbk, 3rd ed, pages 52-53; ASNT NDT Hdbk Vol 10, 3rd ed, page 113

53. What do you call testing with mirrors, borescopes, cameras, and video probes?

- a. Direct visual testing
- b. Indirect visual testing
- c. Augmented visual testing
- d. Isolated visual testing

ASNT VT Hdbk 3rd ed, page 13 left col; ASNT NDT Hdbk Vol 10, 3rd ed, page 48, left col, and 69 "Remote"

54. A radiographer is exposed to radiation for 2 hours at a rate of 50 mR/hr. He then doubles his distance from the source and works another 8 hours. What is his total exposure?

- a. 100 mR
- b. 200 mR
- c. 500 mR
- d. 1700 mR

55. One Ci is the equivalent of _____.

- a. 37 GBq
- b. 37×10^9 Bq
- c. 3.7×10^{10} Bq
- d. 3.7×10^{10} Atomic Disintegrations/s
- e. All of the above

56. Which of the following is not a source of neutrons?

- a. Nuclear reactor
- b. High energy particle accelerator
- c. Co60
- d. Cf252

BASIC LEVEL III PRACTICE EXAM

57. Convert 2 ft-c to lux.

- a. 10.25 lx
- b. 20 lx
- c. 21.5 lx
- d. 0.185 lx

58. A source emits 500 mR/hr per Ci at 1 ft, what would be the exposure rate from a 10 Ci source at 100 ft?

- a. 0.5 mR/hr
- b. 50 mR/hr
- c. 500 mR/hr
- d. 5000 mR/hr

59. A penetrant is said to have good wetting ability when the contact angle is:

- a. Less than 90 degrees
- b. Equal to 90 degrees
- c. More than 90 degrees
- d. Wetting ability is not affected by contact angle

60. A trainee trying to qualify for Level I in penetrant shall have:

- a. 12 hours of training, and 400 hours of NDT experience of which a minimum of 210 hours must be in penetrant
- b. Four hours of training and 400 hours of NDT experience of which a minimum of 210 hours must be in penetrant
- c. 8 hours of training and 130 hours of NDT experience of which a minimum of 70 hours must be in penetrant
- d. Four hours of training, and 130 hours of NDT experience of which a minimum of 70 hours must be in penetrant

SNT-TC-1A Table 6.3.1 A and CP-189 Appendix A

61. Leakage Measurement is defined as

- a. Qualitative Measurement
- b. Quantitative Measurement
- c. A Combination of both Qualitative and Quantitative Measurement
- d. The total leakage

62. An undesirable defect in Sintered Ceramic material used as a structure would be.

- a. Porosity
- b. Variations of density
- c. Variations of Velocity
- d. Changes in Attenuation.

N192-93, last/first; Old 137, right col, last

63. In normal SI measurements, the little letter ie. (m-milli) and the big letter ie. (M-Mega) are correlate to a magnitude difference of _____?

- a. 2
- b. 3
- c. 6
- d. 9

64. Certification is

- a. Written testimony of qualification of a specific NDT method
- b. Skill, knowledge, and successful completion of exams
- c. Average scores of 80% or better, with minimums of 70% on each exam
- d. Skill, knowledge, and demonstrated capability to carry out NDT tasks

SNT-TC-1A 2.1.2

BASIC LEVEL III PRACTICE EXAM

65. One foot candle is equal to _____

- a. 1 lux
- b. 3.28 lux
- c. 1076 lux
- d. 10 lux

66. What tool has a U-shape connected to an anvil?

- a. Micrometer
- b. Surface contour gauge
- c. Vernier Calipers
- d. Dial Indicator

67. What is used in a casting to prevent segregation while the metal is solidifying?

- a. Riser, Runner, or Feeder
- b. Chill
- c. Sprue

68. If the density changes across the entire weld, it is said to be?

- a. Under fill
- b. Concave Root
- c. Undercut
- d. Overlap

69. A storage tank was used in service to contain different types of oils for the past 15 years. If Fluorescent PT was used to check the inside of the welds and the HAZ area, what weld defect is likely to be found?

- a. Creep
- b. Hydrogen Induced Cracking
- c. MIC
- d. Fatigue

API 570 API 653 – *Weld Defect*

70. The input for a portable MT unit while using prods is typically _____.

- a. 120 VAC, Single phase
- b. 120 VAC, 3 phase
- c. 240 HWD C
- d. 120 FWDC

71. EDM is used in NDT for _____.

- a. Manufacture of natural discontinuities in calibration standards
- b. Polishing scanned areas to enhance the inspection environment
- c. Positive material identification
- d. Manufacture of notches and grooves

72. To verify the presence of a crack radiating from a fastener hole the best method would be _____.

- a. Leak detection of the hole
- b. Ultrasonic testing using a longitudinal wave long the hole
- c. Radiography with the beam parallel to the skin
- d. An eddy current pancake probe scanned around the hole

73. A porous magnesium fillet is to be processed by penetrant. What would be the best choice?

- a. Visible dye, water washable
- b. Fluorescent dye, water washable
- c. Visible dye, solvent removable
- d. Fluorescent dye, post-emulsifiable hydrophilic

BASIC LEVEL III PRACTICE EXAM

74. How should AE, audible sound, and UT be placed in order of increasing frequency?
- a. UT, AE, audible sound
 - b. Audible sound, AE, with the lower portion of the UT range overlapping the upper portion of AE
 - c. Audible sound, AE, and UT
 - d. Audible sound, with the lower portion of AE overlapping the upper portion of audible sound, and the upper portion of AE overlapping the lower portion of UT

75. What is the fill factor for a 10 mm coil placed in a 15 mm tube?

- a. .1
- b. .667
- c. 1.5
- d. 0.444

76. What is the fill factor for a 10 mm coil encircling a 9 mm rod?

- a. 0.9
- b. 1.11
- c. 0.81
- d. 0.09

77. Comparing placement of transducers for composites and steel parts, which of the following is true?

- a. Closer spacing for composites
- b. Closer spacing for steel
- c. Random placement for composites
- d. Inner and outer diameters for composites

78. A _____ purge is used with to shield the outer chamber from the atmosphere (the atmosphere may contain trace elements of the tracer gas) during vacuum leak testing.

- a. Helium
- b. Oxygen
- c. Hydrogen
- d. Nitrogen

<https://www.cincinnati-test.com/nitrogen-purge-leak-testing.html>